

* Quartiles

easy

1) 5, 7, 9, 11, 13

$$Q_1 = 6$$

$$Q_2 = 9$$

$$Q_3 = 12$$

$$IQR = 6$$

$$\text{Lower} = Q_1 - 1.5 * IQR$$

$$= 6 - 1.5 * (6)$$

$$= 6 - 9$$

$$= \boxed{-3}$$

$$IQR = Q_3 - Q_1$$

$$12 - 6$$

$$= \boxed{6}$$

$$\text{upper} = Q_3 + 1.5 * IQR$$

$$= 12 + 1.5 * 6$$

$$= 12 + 9$$

$$= \boxed{21}$$

2) 10, 20, 30, 40, 50

$$Q_1 = 15$$

$$Q_2 = 30$$

$$Q_3 = 45$$

$$IQR = 30$$

$$\text{Lower} = Q_1 - 1.5 * (IQR)$$

$$= 15 - 1.5 * (30)$$

$$= 15 - 45$$

$$= \boxed{-30}$$

$$IQR = Q_3 - Q_1$$

$$45 - 15$$

$$= \boxed{30}$$

$$\text{upper} = Q_3 + 1.5 * (IQR)$$

$$= 45 + 1.5 * (30)$$

$$= 45 + 45$$

$$= \boxed{90}$$

3) 10, 10, 10, 10, 10

$$Q_1 = 10$$

$$Q_2 = 10$$

$$Q_3 = 10$$

$$IQR = 0$$

$$Q_1 - 1.5(IQR)$$

$$= 10 - 1.5(0)$$

$$= 10 - 0$$

$$= \boxed{10}$$

$$Q_3 + 1.5(IQR)$$

$$= 10 + 1.5(0)$$

$$= \boxed{10}$$

medium

4) 15, 18, 20, 22, 25, 30

$$Q_1 = 16$$

$$Q_2 = 21$$

$$Q_3 = 27.5$$

$$IQR = 11.5$$

$$Q_1 - 1.5^*(IQR)$$

$$= 16 - 1.5^*(11.5)$$

$$= 16 - 17.25$$

$$= \boxed{-1.25}$$

$$IQR = Q_3 - Q_1$$

$$= 27.5 - 16$$

$$= \boxed{11.5}$$

$$Q_3 + 1.5^*IQR$$

$$= 27.5 + 1.5^*11.5$$

$$= 27.5 + 17.25$$

$$= \boxed{44.75}$$

Hard Level

5) 12, 15, 18, 21, 24, 27, 90

$$Q_1 = 13.5$$

$$Q_2 = 21$$

$$Q_3 = 58.5$$

$$IQR = 45$$

$$\begin{aligned} IQR &= Q_3 - Q_1 \\ &= 58.5 - 13.5 \\ &= 45 \end{aligned}$$

$$\begin{aligned} \text{Lower} &= Q_1 - 1.5 * IQR \\ &= 13.5 - 1.5 * 45 \\ &= 13.5 - 67.5 \\ &= -54 \end{aligned}$$

$$\begin{aligned} \text{Upper} &= Q_3 + 1.5 * IQR \\ &= 58.5 + 1.5 * 45 \\ &= 58.5 + 67.5 \\ &= 126 \end{aligned}$$

Interview level

6) 1, 2, 3, 4, 5, 6, 7, 8, 9

$$Q_1 = 2.5$$

$$Q_2 = 5$$

$$Q_3 = 7.5$$

$$IQR = 5$$

$$\begin{aligned} \text{Lower} &= Q_1 - 1.5 * IQR \\ &= 2.5 - 1.5 * 5 \\ &= 2.5 - 7.5 \\ &= -5 \end{aligned}$$

$$\begin{aligned} IQR &= 7.5 - 2.5 \\ &= 5 \end{aligned}$$
$$\begin{aligned} \text{Upper} &= Q_3 + 1.5 * IQR \\ &= 7.5 + 7.5 \\ &= 15 \end{aligned}$$